

Supplementary Movies for “Vesicle dynamics in confined steady and harmonically modulated Poiseuille flows”

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(Dated: April 3, 2018)

1. *Centred snaking shape of a confined vesicle in steady Poiseuille flow.*

Capillary number and confinement are fixed to $C_k = 1.37$ and $C_n = 0.55$, respectively (corresponds to point B in Fig. 1 of the article). The membrane displays a purely oscillatory tank-treading (green line added for illustration). The position of the vesicle's center of mass is indicated by the black dot. All variables are scaled as described in Sec. II of the article.

2. *Shape transition from centred snaking to the unconfined slipper via the parachute shape under a harmonically modulated Poiseuille flow.*

In a harmonically modulated Poiseuille flow, the centered snaking shape of point B can be forced to evolve to the parachute shape with $\varepsilon_m = 1$. After imposing a steady Poiseuille flow again the vesicle takes on the shape of an unconfined slipper. All variables are scaled as described in Sec. II of the article.

3. *Centred snaking motion with a constant and varying envelope under a harmonically modulated Poiseuille flow.*

In a modulated Poiseuille flow the centered snaking shape of point B displays an oscillation with a constant envelope of the center of mass for $\varepsilon_m = 0.7$ and $f_m = 0.092$. When the modulation frequency is only slightly increased to $f_m = 0.093$ (ε_m fixed), the envelope becomes unstable. The video shows the simulations corresponding to Fig. 6(c) and Fig. 6(d) of the article. All variables are scaled as described in Sec. II of the article.